Joseph Thomas

+1-2404133452 josephthomas2222@gmail.com josephthomas josephgit10

Education

University of Maryland - College Park, MD, USA **May 2024**
Master of Engineering in Robotics *GPA: 3.38/4.0*

Maharashtra Institute of Technology - Pune, MH, India **May 2022**
Bachelor of Technology in Mechanical Engineering (Robotics & Automation) *GPA: 3.5/4.0*

Technical Skills

Programming Languages: Python, C++, MATLAB, SQL

Machine Learning: Deep Learning, Reinforcement Learning, Time-Series Forecasting, Computer Vision, Regression, Classification, Clustering, Optimization Algorithms

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, FastAPI
MLOps & Cloud: Docker, Git, CI/CD, AWS, GCP, Azure, Cloud Architecture
Data Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly

Professional Experience

Health4theWorld **Feb 2025 – July 2025**
AI Applications Developer *Carmel, CA, USA*

- Benchmarked and optimized 5+ large-scale machine learning models for deployment, tuning performance to meet specific accuracy, latency, and compute requirements.
- Enhanced LLM contextual accuracy by customizing OpenWebUI's RAG pipeline to create a specialized knowledge base from the organization's expert medical lectures.
- Engineered a scalable MLOps pipeline on GCP, deploying multiple LLMs using Docker and T4 GPUs. This improved response accuracy by an estimated 20% by integrating medical journals through a RAG pipeline, preparing the system for a future global launch to medical professionals.

Void Robotics **Aug 2024 – Jan 2025**
Robotics Software Engineer *Marathon, FL, USA*

- Optimized robot perception by implementing and fine-tuning computer vision-based SLAM algorithms, significantly improving navigation accuracy in dynamic environments.
- Developed a reusable data validation and automated testing framework to continuously monitor model inputs, improving the quality of training data by 30%.
- Automated configuration management scripts to streamline parameter updates across deployment environments, reducing manual intervention.
- Validated system performance through extensive testing with ROS2/Gazebo simulations, achieving 99.8% uptime across 100+ test runs to ensure real-time reliability.

Select Projects

End-to-End Demand Forecasting System for Supply Chain Logistics **May 2025 – Jun 2025**

- Orchestrated a daily MLOps workflow with Apache Airflow to automate data preprocessing, model training, and batch inference.
- Engineered the full system using Docker, exposing predictions via a FastAPI microservice for on-demand inference.
- Implemented and tuned multiple models (Prophet, LSTM, XGBoost), using Optuna for hyperparameter search to achieve a 15% reduction in RMSE.

Enhancing Efficiency in Hate-Speech Detection **Sep 2023 – Jan 2024**

- Developed an end-to-end ML pipeline for hate-speech detection on resource-constrained devices by integrating advanced data preprocessing, feature engineering, and model training. Leveraged knowledge distillation with TinyBERT and DistilBERT to achieve a 25× reduction in model size and up to 37.85× faster prediction times while maintaining high accuracy.
- Demonstrated model generalization with up to a 54.65% improvement in F1 score and a 30.50% increase in accuracy on the ToxiGen dataset compared to BERT, ensuring effective detection across diverse contexts.

Vehicle Speed Estimation using Optical Flow **Apr 2023 – May 2023**

- Developed a real-time computer vision pipeline for vehicle speed estimation, deploying a YOLO-based object detection model on embedded hardware (Raspberry Pi 4).
- Architected a vision pipeline to stream 30 FPS video from a Raspberry Pi camera to a laptop for processing, evaluating multiple optical flow algorithms (Lucas-Kanade, RAFT) for speed estimation.